

What is a codec?

The audio files basically have only binary digits and this cannot be read directly by the transmission device for playback. A program needs to read the file and decode it in a language which can be understood by the output device. The difference between a codec and DAC is that a DAC converts the digital audio stream into analogue signals for the drivers, and every DAC will have a codec which reads the file and converts the file into an audio stream which is then converted to analogue signals by the DAC. Codecs can be lossy or lossless. Lossy codecs are very fast and consume lesser processing power than lossless codecs, because they trade accuracy for speed. Lossless audio is used in cases where there is a requirement for a raw audio stream.

Audio File Types

The rise of streaming and online music playback has virtually eliminated the need to worry about audio file formats. But in a world here audiophiles obsess over the smallest details, file formats are one of the most important factors to get great sound. Audio file formats extend beyond merely giving an extension to the file name. The file format dictates the quality of the file and the compression used. File formats can be thought about as containers for storage of data about the sound streams.



We are spoilt for choice when it comes to audio formats

MP3

MP3 is the most common file format used for audio. MP3 or MPEG-1 and/or MPEG-2 Audio Layer III has become the standard for audio files. The file format was developed by the Moving Picture Experts Group (MPEG). This is a type of lossy compression, which means, some sounds are removed to reduce the file size. The removal of the sounds is determined by the algorithm used. Most often, there are thresholds set based on the generic hearing capacity of humans. Any sounds above the maximum threshold and below the minimum threshold are removed. The reason for the rise in the power of MP3 lies with the problems caused by CDs. CDs were considered to be the pinnacle of the digital audio revolution, but they were bulky, hard to carry and susceptible to damage and loss. And anyone with a large music